

Exact Codimension-Two Reaching Thresholds for Binary Completely Reachable Automata

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Abstract

A deterministic finite automaton is completely reachable if every nonempty subset of its state set is the image of the full state set under some word. For an n -state binary completely reachable automaton, we determine the exact worst-case reaching threshold of subsets of size $n - 2$. Writing $F_{\text{BCR}}(n, n - 2)$ for the maximum, over all such automata and all $(n - 2)$ -subsets, of the length of a shortest reaching word, we prove for every $n \geq 4$ that

$$F_{\text{BCR}}(n, n - 2) = \begin{cases} 2n, & n \text{ odd,} \\ \max\{2n, \frac{5}{2}n - 3\}, & n \text{ even.} \end{cases}$$

The upper bound is self-contained after an elementary structural reduction: every binary completely reachable automaton with at least three states has one defect-one letter and one cyclic permutation letter. All non-antipodal pairs of missing states are reached within $2n$ steps. The only obstruction is an even-state antipodal configuration, for which a two-route argument through the two preimages of the duplicated state yields the sharper bound $5n/2 - 3$. The lower bound $2n$ is supplied by the classical Černý family, while Zhu's even-state construction supplies $5n/2 - 3$ for every even $n \geq 10$. The remaining case $n = 8$ is certified by exhaustive enumeration of all 141,120 normalized defect-one maps: 136,704 are completely reachable and the exact codimension-two maximum is 17, attained by 68 automata. Thus the previously known gap of three between the general $5n/2$ upper bound and the binary $5n/2 - 3$ lower bound is closed for binary completely reachable automata.

1 Introduction

A deterministic finite automaton (DFA) $\mathcal{A} = (Q, \Sigma)$ is *completely reachable* if for every nonempty subset $S \subseteq Q$ there exists a word $w \in \Sigma^*$ such that $Qw = S$. The length of a shortest such word is the *reaching threshold* of S . Complete reachability is closely related to synchronization and to subset-reachability questions surrounding Don's conjecture [4].

For arbitrary completely reachable automata, Ferens and Szykuła proved the upper bound

$$2n(n - |S|) - nH_{n-|S|}$$

for the reaching threshold of a nonempty proper subset S in an n -state automaton. For $|S| = n - 2$ this is $5n/2$ [3]. Zhu constructed an infinite family of binary completely reachable automata, for even $n \geq 10$, containing an $(n - 2)$ -subset whose reaching threshold is at least $5n/2 - 3$ [5]. Ferens and Szykuła consequently described the general codimension-two upper bound as almost tight in view of this binary lower bound [3].

This note determines the exact binary value. The proof isolates a single obstruction to the direct $2n$ construction: the duplicated state must be the unique nonzero element of order two, and the target must omit an antipodal pair. In that case two complementary routes through the preimages of the duplicated state force the $5n/2 - 3$ upper bound.

2 Definitions

Words act on the right. For a subset $S \subseteq Q$ and word w , write $Sw = \{qw : q \in S\}$. For a reachable nonempty subset S , define

$$\text{rt}_{\mathcal{A}}(S) = \min\{|w| : Qw = S\}.$$

Let BCR_n denote the class of n -state binary completely reachable DFAs, and define

$$F_{\text{BCR}}(n, n-2) = \max_{\mathcal{A} \in \text{BCR}_n} \max_{\substack{S \subseteq Q \\ |S|=n-2}} \text{rt}_{\mathcal{A}}(S).$$

For a transformation $a : Q \rightarrow Q$, its defect is $|Q| - |Qa|$. A defect-one transformation has one excluded state, denoted $\text{excl}(a)$, and one duplicated image state, denoted $\text{dupl}(a)$; the duplicated state has exactly two preimages, while every other image state has exactly one.

3 The binary structural reduction

We use the following elementary reduction, included for completeness. It is consistent with the structural analysis of binary completely reachable automata by Casas and Volkov [1].

Lemma 1 (Cycle-defect-one form). *Let $\mathcal{A} = (Q, \{a, b\})$ be completely reachable with $n = |Q| \geq 3$. After possibly exchanging the letters, a has defect one and b acts as a single n -cycle.*

Proof. Since every $(n-1)$ -subset is reachable, some letter has rank $n-1$; call it a . If b were also singular, every nonempty word ending in a or b would have image contained in Qa or Qb , respectively. Hence at most two subsets of size $n-1$ could be reached, contradicting complete reachability for $n \geq 3$. Thus b is a permutation.

Suppose that b has more than one cycle. Let e be the excluded state of a and let C be the b -cycle containing e . Then $C \not\subseteq Qa$. By complete reachability choose a shortest word w with $Qw = C$. If $w = ub$, then $Qu = Cb^{-1} = C$, because C is b -invariant, contradicting minimality. If $w = ua$, then $Qw \subseteq Qa$, contradicting $C \not\subseteq Qa$. Hence b is a single n -cycle. $\square$

By relabeling states, with no change in word lengths, we may therefore assume throughout that

$$Q = \mathbb{Z}_n, \quad qb = q + 1 \pmod{n}, \quad \text{excl}(a) = 0.$$

Write $d = \text{dupl}(a) \in \mathbb{Z}_n \setminus \{0\}$.

4 Upper bound

The next observation handles every target except one even-state configuration.

Lemma 2 (Direct two-hole construction). *Let $S = Q \setminus \{x, y\}$ and put $z = y - x \pmod{n}$. If $z \neq d$, then S has a reaching word of length at most $2n$. The same holds if $-z \neq d$.*

Proof. Assume $z \neq d$. Since $z \neq 0$ and z is not the duplicated state, it has a unique preimage i under a , so $ia = z$. Then

$$Q \xrightarrow{a} Q \setminus \{0\} \xrightarrow{b^i} Q \setminus \{i\} \xrightarrow{a} Q \setminus \{0, z\} \xrightarrow{b^x} Q \setminus \{x, y\}.$$

Thus $ab^i ab^x$ reaches S . Choosing the standard representatives $i, x \in \{0, \dots, n-1\}$ gives length $i + x + 2 \leq 2n$. If $-z \neq d$, exchange x and y and apply the same argument. $\square$

The direct construction can fail in both orientations only when $d = -d \neq 0$. Thus $n = 2m$, $d = m$, and the missing pair is antipodal. We now handle exactly this case.

Let

$$a^{-1}(m) = \{r, s\}, \quad \Delta = s - r \pmod{n}.$$

Lemma 3 (The preimage gap is not antipodal). *In a binary completely reachable automaton in the above antipodal configuration, $\Delta \neq m$.*

Proof. Assume for contradiction that $\Delta = m$. Define a new letter $c = b^r a$. Since b^n is the identity transformation,

$$c = b^r a, \quad a = b^{n-r} c,$$

so $\langle a, b \rangle = \langle c, b \rangle$. Hence the automaton $\mathcal{A}' = (Q, \{b, c\})$ has the same transition monoid and is completely reachable whenever the original automaton is.

We have $Qc = Qa = Q \setminus \{0\}$. Moreover,

$$qc = m \iff (q + r)a = m \iff q + r \in \{r, s\},$$

so under $s - r = m$,

$$c^{-1}(m) = \{0, m\}.$$

For $i \in \{0, \dots, m-1\}$ let

$$T_i = Q \setminus \{i, i + m\}.$$

Because $\mathcal{A}'$ is completely reachable, every T_i is reachable. Choose, among all words over $\{b, c\}$ reaching any T_i , one of minimum length, say $Qw = T_i$. The word is nonempty.

If $w = ub$, then

$$Qu = T_i b^{-1} = T_{i-1},$$

where the subscript is taken modulo m . This contradicts the minimality of w .

If $w = uc$, put $P = Qu$. Since $Pc = T_i \subseteq Qc = Q \setminus \{0\}$, the set T_i cannot contain 0, so necessarily $T_i = T_0 = Q \setminus \{0, m\}$. Also $m \notin Pc$ and $c^{-1}(m) = \{0, m\}$, hence $P \cap \{0, m\} = \emptyset$ and therefore $P \subseteq T_0$. Since a transformation cannot increase cardinality,

$$|P| \geq |Pc| = n - 2 = |T_0|.$$

Thus $P = T_0$, so the shorter word u already reaches an antipodal complement, again contradicting minimality.

Both possible final letters lead to contradictions. Therefore $\Delta \neq m$. $\square$

Lemma 4 (Antipodal detour). *If $n = 2m$, $d = m$, then $Q \setminus \{0, m\}$ has a reaching word of length at most $2n - 2$.*

Proof. We have $\Delta \neq 0$ because $r \neq s$, and $\Delta \neq m$ by Lemma 3. Also $\Delta \neq -\Delta$, because $2\Delta = 0$ in $\mathbb{Z}_{2m}$ would imply $\Delta \in \{0, m\}$. Hence Δ and $-\Delta$ are distinct simple image states of a . Let

$$ia = \Delta, \quad ja = -\Delta$$

be their unique preimages.

The four states i, j, r, s are pairwise distinct: $i \neq j$ because their images differ, while neither i nor j can equal r or s because $ra = sa = m$.

Consider

$$u = ab^i ab^r a, \quad v = ab^j ab^s a.$$

For u , the missing-state sets evolve as

$$\{0\} \rightarrow \{i\} \rightarrow \{0, \Delta\} \rightarrow \{r, s\} \rightarrow \{0, m\},$$

so $Qu = Q \setminus \{0, m\}$. The word v does the same using $-\Delta$.

Using representatives $i, j, r, s \in \{0, \dots, n-1\}$, pairwise distinctness gives

$$i + j + r + s \leq (n-1) + (n-2) + (n-3) + (n-4) = 4n - 10.$$

Therefore

$$\min\{i + r, j + s\} \leq 2n - 5.$$

At least one of u, v consequently has length at most $3 + (2n - 5) = 2n - 2$. □

Proposition 5 (Binary codimension-two upper bound). *For every $n \geq 4$,*

$$F_{\text{BCR}}(n, n-2) \leq \begin{cases} 2n, & n \text{ odd}, \\ \max\{2n, \frac{5}{2}n - 3\}, & n \text{ even}. \end{cases}$$

Proof. Lemma 2 gives the bound $2n$ unless both oriented differences of the missing pair equal d . This can happen only when $n = 2m$, $d = m$, and the pair is antipodal. By Lemma 4, the base pair $\{0, m\}$ is reached within $2n - 2$. Every antipodal pair is $\{t, t + m\}$ for a unique $0 \leq t \leq m - 1$, so appending b^t yields length at most

$$2n - 2 + (m - 1) = \frac{5}{2}n - 3.$$

□

5 Matching lower bounds

The lower bounds come from three sources.

First, the classical Černý automata [2] are completely reachable and attain Don's bound for every subset size; in particular they contain $(n-2)$ -subsets of reaching threshold $2n$ [3]. Hence

$$F_{\text{BCR}}(n, n-2) \geq 2n \quad (n \geq 4).$$

Second, Zhu constructed, for every even $n \geq 10$, a binary completely reachable automaton A_n with an $(n-2)$ -subset of reaching threshold at least

$$\frac{5}{2}n - 3$$

[5]. We use this result only in its explicit even-state range.

The sole remaining lower-bound case is $n = 8$. We certify it exhaustively.

Proposition 6 (Exact finite certificate at $n = 8$). *For $n = 8$, $F_{\text{BCR}}(8, 6) = 17$.*

Computer-assisted proof. Fix $b : q \mapsto q + 1 \pmod{8}$ and normalize the defect-one letter a to have excluded state 0. Every such map is generated by choosing the duplicated image $d \in \{1, \dots, 7\}$, its two preimages, and a bijection from the remaining six domain states to the remaining six image states. Thus the search contains exactly

$$7 \binom{8}{2} 6! = 141,120$$

defect-one maps.

For each map, the verifier performs a direct breadth-first search on the $2^8 - 1 = 255$ nonempty subsets under the two letters; no structural recognition theorem is used. Exactly 136,704 automata are completely reachable. Among them the largest shortest reaching distance to a six-state subset is 17, attained by 68 automata. One normalized witness is

$$a = (5, 2, 3, 6, 4, 4, 7, 1),$$

with hard subset $\{0, 1, 2, 4, 5, 6\}$. The accompanying verifier exhausts the full normalized search space and asserts all quoted counts. $\square$

6 Exact threshold

Theorem 7 (Exact codimension-two threshold). *For every integer $n \geq 4$,*

$$F_{\text{BCR}}(n, n-2) = \begin{cases} 2n, & n \text{ odd}, \\ \max\{2n, \frac{5}{2}n - 3\}, & n \text{ even}. \end{cases}$$

Equivalently, the value is $2n$ for odd n and for $n = 4, 6$, while it is $5n/2 - 3$ for even $n \geq 8$.

Proof. Combine Proposition 5 with the Černý lower bound, Proposition 6, and Zhu's lower bound for even $n \geq 10$. $\square$

7 Relation to previous bounds

Ferens and Szykuła's general completely reachable bound gives, for $|S| = n - 2$,

$$2n \cdot 2 - nH_2 = 4n - \frac{3}{2}n = \frac{5}{2}n$$

[3]. Their 2026 paper explicitly notes Zhu's binary lower bound $5n/2 - 3$ and describes the codimension-two upper bound as almost tight. Theorem 7 shows that, within the binary completely reachable class, the additive gap of three is exact and disappears.

The proof also explains why the larger value is confined to even n : the direct $2n$ route fails in both orientations only when the duplicated state is its own additive inverse, which is exactly the nonzero order-two element of $\mathbb{Z}_n$.

8 Reproducibility

The repository accompanying this paper contains the script `verify_n8.py`. It uses only the Python standard library. Running

```
python verify_n8.py --n 8 --json verification_n8.json
```

enumerates all 141,120 normalized defect-one maps and independently recomputes complete reachability and all codimension-two distances by subset BFS. The expected audit summary is:

normalized defect-one maps	141,120
completely reachable	136,704
maximum codimension-two threshold	17
extremal automata	68

9 Conclusion

The exact second-from-top subset reaching threshold for binary completely reachable automata is governed by a simple dichotomy. A generic pair of missing states is created directly within $2n$ steps. The only obstruction is antipodal and therefore even-state; a self-contained detour argument then gives $5n/2 - 3$, matching the known lower construction. The $n = 8$ boundary case is independently exhausted by finite computation.

References

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